increase of 10 degrees Celsius would increase the weight of a one-pound cube of gold by about a millionth of a billionth of a pound, so the effect is minuscule), and with this solution you win release from the Hall.

Well, almost. Because your crime was particularly devious, at the last minute the parole board decides that you must solve a second puzzle. You are given two identical old-time Jack-in-the-box toys, and your new challenge is to find a way to make each have a different weight. But in this go-around, not only are you forbidden to change the amount of mass in either object, you are also required to keep both at exactly the same temperature. Again, were Newton given this puzzle, he would immediately resign himself to life in the Hall. Since the toys have identical masses, he would conclude that their weights are identical, and so the puzzle is insoluble. But once again, your knowledge of general relativity comes to the rescue: On one of the toys you compress the spring, tightly squeezing Jack under the closed lid, while on the other you leave Jack in his popped-up posture. Why? Well, a compressed spring has more energy than an uncompressed one; you had to exert energy to squeeze the spring down and you can see evidence of your labor because the compressed spring exerts pressure, causing the toy's lid to strain slightly outward. And, again, according to Einstein, *any* additional energy affects gravity, resulting in additional weight. Thus, the closed Jack-in-the-box, with its compressed spring exerting an outward pressure, weighs a touch more than the open Jack-in-the-box, with its uncompressed spring. This is a realization that would have escaped Newton, and with it you finally *do* earn back your freedom.

The solution to that second puzzle hints at the subtle but critical feature of general relativity that we're after. In his paper presenting general relativity, Einstein showed mathematically that the gravitational force depends not only on mass, and not only on energy (such as heat), but also on any *pressures* that may be exerted. And this is the essential physics we need if we are to understand the cosmological constant. Here's why. Outward-directed pressure, like that exerted by a compressed spring, is called *positive pressure.* Naturally enough, positive pressure makes a positive contribution to gravity. But, and this is the critical point, there are situations in which the pressure in a region, unlike mass and total energy, can be *negative,* meaning that the pressure sucks inward instead of pushing outward. And although that may not